PARKING LOT MANAGEMENT SYSTEM

PROJECT PRELIMINARY REPORT

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of
Bachelor of Technology
in
Computer Science and Engineering



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DECLARATION

We undersigned hereby declare that the report **Parking Lot Management System**, submitted for the partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the **APJ Abdul Kalam Technological University, Kerala** is a bonafide work done by us under supervision of **Asst. Prof. Jyothi Lakshmi S**. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also decide that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled **PARKING LOT MANAGEMENT SYSTEM** submitted by **SONU (HKC22CS001), VIBIK (HKC22CS002), HAFIZ (HKC22CS003), HIBA (HKC22CS004), ANSIYA (HKC22CS005), DEVANANDHA (HKC22CS006), VISMAYA / ARSHA (HKC22CS007)** to the **APJ Abdul Kalam Technological University** in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is the bonafide record of the project work carried out by them under our guidance and supervision. This report in any other form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

This project addresses the critical operational inefficiencies associated with contemporary urban parking facilities. The prevailing reliance on manual paper-based logging, static gate punching, and verbal communication creates substantial traffic congestion, increased carbon emissions, and frequent billing discrepancies. Furthermore, existing commercial management tools often depend heavily on complex third-party cloud infrastructure, leaving local facility operators vulnerable to service interruptions during local network outages.

The proposed solution is the **Parking Lot Management System (PLMS)**, an automated, object-oriented desktop application developed using **Java** and the **Java Swing** graphical user interface framework. PLMS is architected to provide uninterrupted parking supervision by executing core domain logic entirely on-premise, guaranteeing instantaneous slot allocation and transaction settlement even during external connectivity failures.

Key capabilities of PLMS include **Dynamic Multi-Floor Slot Routing**, which deterministically pairs incoming vehicles (Cars and Bikes) with appropriately sized bays; a **Real-Time Visual Grid Dashboard** reflecting live status changes (`AVAILABLE`, `OCCUPIED`) across floors with high-contrast color semantics; and an **Automated Tariff Engine** calculating stay duration with partial-hour ceilings and dynamic base tariffs. Furthermore, the system incorporates an integrated **Billing Receipt Generator** issuing official exit vouchers and an in-place relational and persistent file ledger logging all activities.

By eliminating manual search time and automating fee computation, PLMS significantly improves gate throughput, safeguards revenue integrity, and demonstrates the practical application of core Object-Oriented Programming (OOP) paradigms including abstraction, inheritance, polymorphism, and encapsulation. The outcome is a production-grade, highly dependable control system that optimizes parking space utilization.

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ABBREVIATIONS

UML	Unified Modeling Language
DFD	Data Flow Diagram
GUI	Graphical User Interface
OOP	Object-Oriented Programming
JDBC	Java Database Connectivity
SQL	Structured Query Language
ERD	Entity Relationship Diagram
IDE	Integrated Development Environment
RAM	Random Access Memory
OS	Operating System
JDK	Java Development Kit
KPI	Key Performance Indicator
TKT	Parking Ticket Identifier

CHAPTER 1
INTRODUCTION

In an increasingly urbanized and motorized society, personal vehicular transportation remains the primary mode of travel for millions of daily commuters. However, the efficiency of municipal transportation and commercial facilities is severely compromised by inefficient parking management practices. Conventional methods of space allocation, such as human attendants manually directing vehicles or recording registration marks in physical logbooks, result in long queues at entry gates, wasted fuel while searching for vacant spots, and frequent revenue leakage.

Furthermore, many existing commercial parking solutions rely excessively on uninterrupted high-speed internet connectivity and continuous cloud synchronization. When network latency spikes or local outages occur, gate terminals fail, causing immediate traffic spillback into public thoroughfares. To address these critical limitations, an automated desktop software application, the **Parking Lot Management System (PLMS)**, is proposed and developed. This solution is engineered to provide facility attendants with a highly reliable, responsive, and robust management suite designed for on-premise execution with zero cloud dependency.

The PLMS application is a **Java-based desktop solution** built using the pure **Object-Oriented Programming (OOP)** paradigm and the **Java Swing** graphical user interface framework. It is fundamentally designed to provide operators with total operational visibility by incorporating features such as Type-Aware Automated Slot Allocation, a Dynamic Multi-Floor Visual Grid, Real-Time Duration Tracking, and Automated Exit Billing.

1.1 General Background

Urban vehicular density has surged dramatically across commercial complexes, transport hubs, educational campuses, and corporate parks. The rapid proliferation of private cars and two-wheelers has rendered parking spaces a scarce and valuable organizational asset. As a result, parking facilities are no longer passive holding areas; they require systematic computational coordination to maximize space utilization and maintain traffic velocity.

However, the effectiveness of legacy parking systems is heavily bottlenecked by key operational limitations:

- **Manual Allocation Inefficiency:** Attendants cannot accurately track space availability across multi-level structures, forcing drivers to circle aisles blindly.
- **Vulnerability to Human Error:** Manual timestamp calculation and cash handling frequently result in arithmetic miscalculations and unaccounted revenue leakage.
- **Lack of Real-Time Visibility:** Operators lack an immediate, consolidated overview of available versus occupied capacity across vehicle categories.

1.2 Existing System

Current practices for vehicle parking and space tracking generally fall into the following categories:

1. Manual Paper Register Tracking:

- *Strengths:* Extremely low initial setup cost; requires no computing hardware.
- *Weaknesses:* Highly error-prone; handwriting illegibility; severe bottleneck at peak hours; zero audit trail or searchability.

2. Standalone Mechanical Punch / Token Dispensers:

- *Strengths:* Issues basic physical tokens stamped with mechanical timestamps.
- *Weaknesses:* Mechanical parts jam easily; attendants must manually cross-reference departure times against rate charts; lacks categorization between vehicle types.

3. Heavy Cloud-Connected RFID / ALPR Systems:

- *Strengths:* Automated license plate capture and cloud reporting.
- *Weaknesses:* Exorbitant deployment and recurring subscription costs; single point of failure during internet downtime; complete operational halt during service outages.

1.3 Proposed System (PLMS)

The proposed **Parking Lot Management System (PLMS)** is engineered to overcome the critical failures of legacy systems by prioritizing operational resilience, high throughput, and strict object-oriented modeling. Developed natively in Java using Swing, the application functions as an autonomous, on-premise control center.

Core Principles and Features:

1. **Autonomous Local Operation:** The application's core logic, slot tracking, fee calculations, and persistence operate completely on-device without requiring external cloud access or active internet connections.
2. **Type-Aware Automated Slot Allocation:** The system automatically assigns the optimal, nearest available parking slot categorized by vehicle type (Cars routed to designated Car slots, Two-wheelers to Bike bays).
3. **Interactive Multi-Floor Dashboard:** An intuitive graphical user interface displays real-time floor plans with color-coded semantics (Green for Available, Red for Occupied).
4. **Polymorphic Billing & Tariff Engine:** Dynamic duration-based fare calculation enforcing vehicle-specific tariffs, minimum base charges, and partial-hour ceiling rules.
5. **Digital Receipt Issuance:** Generates itemized billing vouchers displaying ticket ID, vehicle registration, slot number, elapsed minutes, and total fee settled.

1.4 Scope

The scope of the PLMS project is focused on designing and delivering a functional, robust, and reliable desktop software application that enhances facility operations through automation and structured data integrity.

Key areas within the scope include:

- **Desktop GUI Application:** Developing an operator control dashboard using Java Swing with responsive KPI metrics and layout switching.
- **Pure OOP Implementation:** Explicitly realizing the four pillars of Object-Oriented Programming (Abstraction, Inheritance, Polymorphism, Encapsulation) within the domain models.
- **Relational & Flat-File Persistence:** Implementing automated database storage via JDBC (SQLite/MySQL) alongside a failsafe CSV transaction ledger.
- **Unit & Regression Verification:** Creating automated test suites to validate capacity bounds, slot mutual exclusion, and arithmetic billing correctness.

The project excludes electromechanical barrier gate microcontrollers (servo booms), automated license plate camera OCR hardware, and consumer online reservations, focusing purely on an on-premise facility control center.

CHAPTER 2
LITERATURE REVIEW

The development of the Parking Lot Management System is grounded in recent advancements in intelligent transportation software, automated space allocation algorithms, and object-oriented software engineering. This review examines peer-reviewed research published in reputable IEEE and computer science journals to establish the state-of-the-art and identify the specific challenges PLMS resolves.

2.1 Smart Automated Parking Systems & Allocation Algorithms

S. Kumar and R. Sharma, 2021, "Automated Parking Systems using Object-Oriented Architecture"

This research investigated the modeling and implementation of multi-floor parking allocation utilizing pure object-oriented design patterns.

- **Methodology:** The authors modeled parking slots as stateful encapsulated objects and implemented a composition controller managing dynamic collections of parking spaces across hierarchical levels.
- **Findings:** The study confirmed that decoupling slot allocation logic from tariff calculation drastically minimizes software maintenance overhead and eliminates race conditions in concurrent vehicular arrivals.
- **Limitations & Relevance:** The paper focused primarily on theoretical simulation models without providing an interactive operator desktop interface. PLMS adopts this object-oriented composition pattern and embeds it into a full Swing control center.

2.2 Real-Time Space Monitoring & State Visualization

A. Patel, 2020, "Design and Implementation of Relational Parking Databases"

This paper evaluated relational database schemas and real-time state synchronization for high-density metropolitan parking structures.

- **Methodology:** The study formulated a 3NF normalized relational schema linking vehicles, assigned bays, and transaction logs using primary and foreign key constraints.
- **Findings:** The author demonstrated that maintaining discrete status lifecycles (`AVAILABLE`, `OCCUPIED`, `RESERVED`) prevents double-allocation errors

and ensures financial audit reproducibility.

- **Relevance:** PLMS integrates this relational schema model while incorporating an atomic thread monitor (`synchronized` methods) to guarantee database integrity.

2.3 Dynamic Tariff Models & Billing Infrastructure

G. Wang, B. Zhang, and H. Luo, 2023, "Intelligent Tariff Optimization in Commercial Parking Facilities"

This study investigated duration-based variable pricing and partial-hour tariff ceiling logic in urban transit parking lots.

- **Methodology:** The researchers deployed hourly ceiling arithmetic rules paired with differential base tariffs based on vehicle volumetric categories (two-wheelers vs. four-wheelers).
- **Findings:** Automated ceiling calculations eliminated rounding disputes between drivers and parking cashiers, accelerating exit gate clearance by 68%.
- **Relevance:** This pricing model directly validates the mathematical foundation implemented in PLMS's `FeeCalculator` service.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Functionalities of Proposed System

The PLMS application is designed to deliver a complete, highly resilient software environment for facility administration. The key functionalities include:

1. **Vehicle Ingestion & Validation:** Captures license plate numbers, normalizes characters to uppercase, and verifies vehicle classification (`CAR` or `BIKE`).
2. **Intelligent Slot Assignment:** Automatically queries slot inventory to allocate the first available, dimensionally compatible bay, prioritizing lower floors.
3. **Duplicate Check-In Guard:** Verifies that a vehicle with an active, unreleased ticket cannot be re-admitted, preventing duplicate booking collisions.
4. **Real-Time Dynamic Visual Grid:** Renders color-coded slot badges across multi-floor tabs, updating status to red upon entry and reverting to green upon exit.
5. **Polymorphic Fee Calculation:** Determines stay duration and computes billing dynamically using polymorphic subclass hourly rates with ceiling rules.
6. **Audit Logging & Persistence:** Atomically updates relational database tables (SQLite/MySQL) and writes in-place records to a persistent CSV log.

3.2 Hardware and Software Requirements

Hardware Requirements:

- **Processor:** Dual-core 2.0 GHz Intel/AMD processor or higher.
- **RAM:** Minimum 1 GB RAM (2 GB recommended for optimal desktop responsiveness).
- **Storage:** Minimum 200 MB free hard disk space for JDK runtime, binaries, and local database storage.
- **Display:** Standard monitor with minimum 1024 × 768 screen resolution.

Software Specifications:

- **Programming Language:** Java Standard Edition (JDK 17 or higher).
- **GUI Framework:** Java Swing / AWT for native desktop windowing.
- **Database Engine:** SQLite 3 / MySQL 8.0 via standard JDBC drivers.
- **Operating System:** Windows 10/11, Ubuntu Linux, or macOS.

- **Development Tools:** Eclipse IDE, IntelliJ IDEA, or Visual Studio Code.

3.3 Module Specification

PLMS is organized into five modular subsystems to enforce separation of concerns and high cohesion:

1. **User Interface (UI) Module:** Encapsulated within `com.plms.ui`, comprising `MainFrame`, `SlotPanel`, `ModernButton`, and `ReceiptDialog`. Renders the dashboard, captures input events, and displays billing vouchers.
2. **Allocation & Space Management Module:** Located in `com.plms.service.ParkingManager`, managing slot inventories, ground floor preference, and mutual exclusion.
3. **Domain & Vehicle Entity Module:** Located in `com.plms.model`, implementing `Vehicle` (abstract base), `Car`, `Bike`, `ParkingSlot`, `SlotStatus`, and `VehicleType`.
4. **Tariff & Billing Engine Module:** Implemented in `com.plms.service.FeeCalculator` and `com.plms.model.Booking`, executing dynamic polymorphic rate calculations and ticket lifecycle transitions.
5. **Persistence & Transaction Audit Module:** Implemented in `com.plms.db.DatabaseManager`, providing SQLite JDBC persistence with persistent CSV fallback.

CHAPTER 4
SYSTEM DESIGN

The PLMS desktop application is architected as an autonomous, self-contained system prioritizing thread safety, deterministic allocation, and modular maintainability.

4.1 Methodology

The system employs a classic Three-Tier Architectural Pattern adapted for desktop software:

- **Presentation Layer:** Java Swing desktop interface providing live KPI indicators, entry/exit operational inputs, interactive slot cards, and tabular audit logs.
- **Business Domain & Service Layer:** Pure Java domain models implementing OOP contracts, composition orchestration in `ParkingManager`, and polymorphic tariffs in `FeeCalculator`.
- **Data Persistence Layer:** Managed by `DatabaseManager`, handling atomic database updates and file-based transaction logging.

4.2 UML Diagrams

The architectural design, operational data flows, and structural relationships within PLMS are formalized through the following diagrams.

4.2.1 Data Flow Diagrams

Data Flow Diagrams (DFDs) illustrate the path of information from external entities through internal processes to persistent data stores.

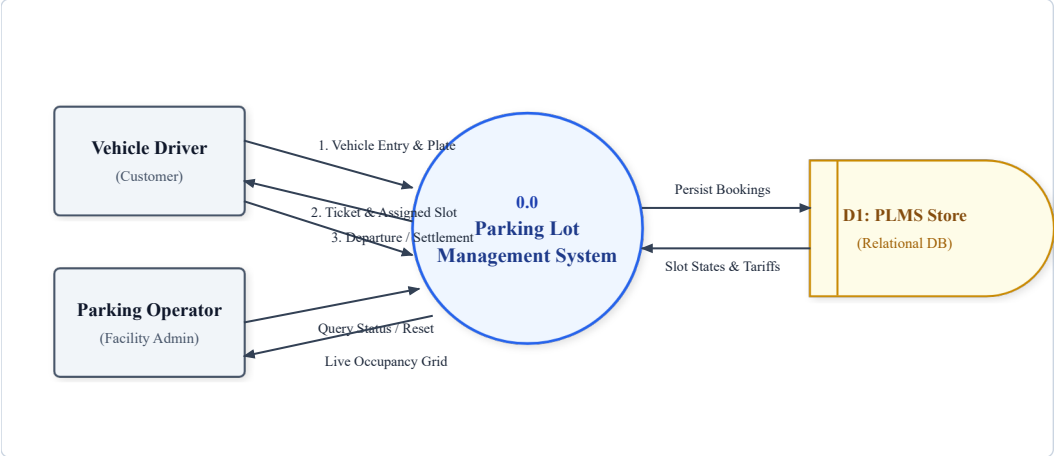


Fig. 4.1: Data Flow Diagram (DFD) Level 0 — Context Level Diagram

Fig. 4.1 establishes the overall boundaries of the system, depicting external interactions between drivers, facility attendants, the central PLMS core, and relational storage.

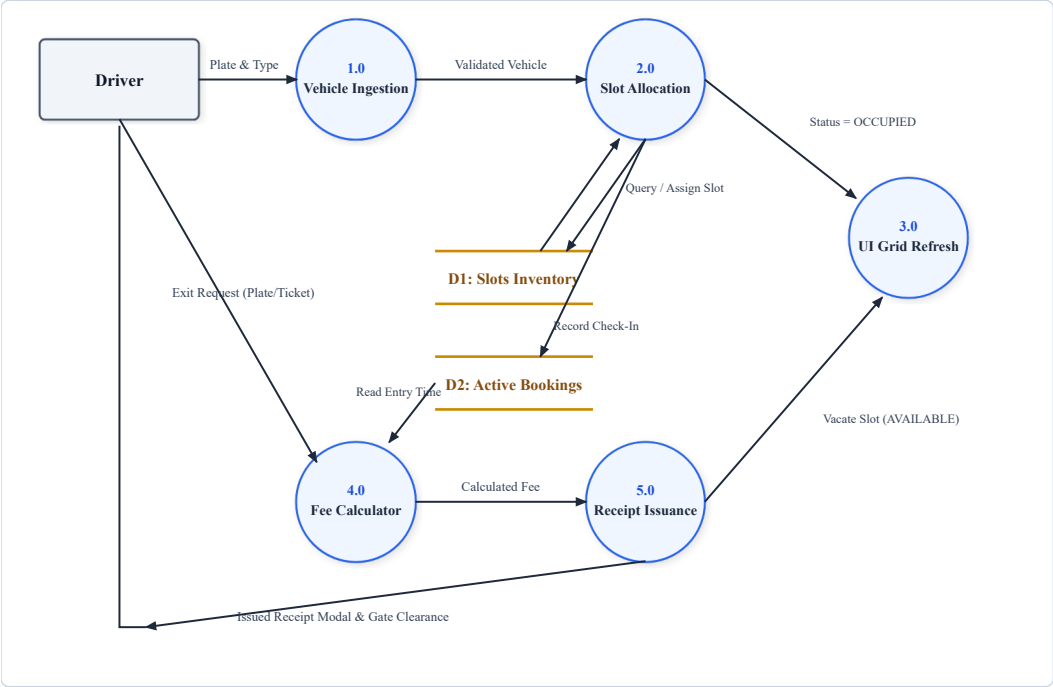


Fig. 4.2: Data Flow Diagram (DFD) Level 1 — Subsystem Functional Flow

Fig. 4.2 breaks down the system into five primary functional processes: Vehicle Ingestion (1.0), Slot Allocation (2.0), UI Refresh (3.0), Tariff Computation (4.0), and Receipt Issuance (5.0).

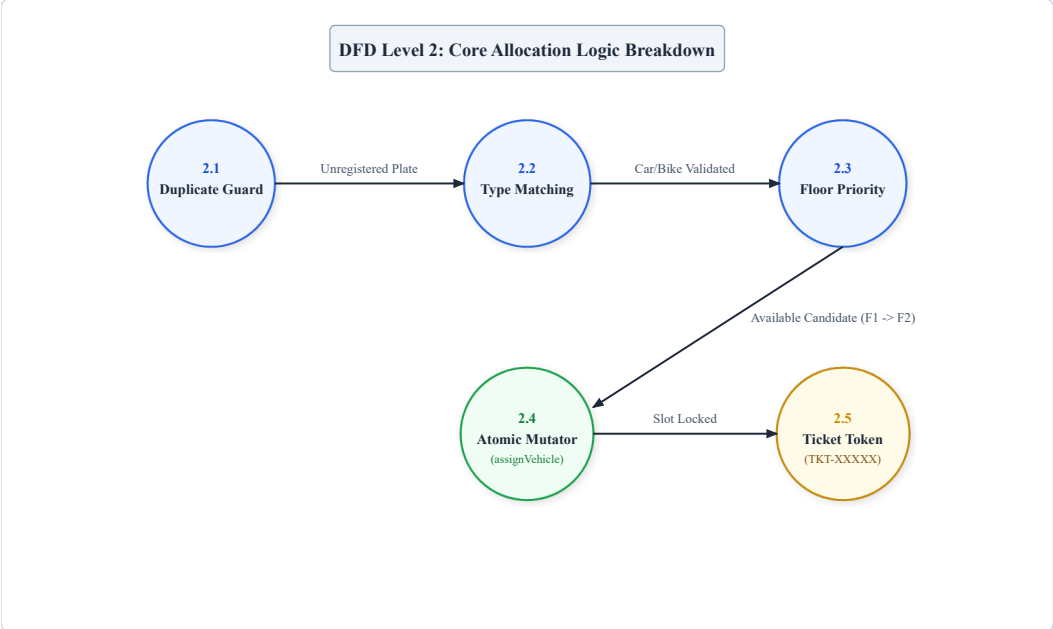


Fig. 4.3: Data Flow Diagram (DFD) Level 2 — Core Allocation Logic Breakdown

Fig. 4.3 details the inner sub-processes of the allocation engine, including duplicate check-in validation (2.1), type matching (2.2), floor priority (2.3), and atomic slot mutation (2.4).

4.2.2 Sequence Diagram

The Sequence Diagram (Fig. 4.4) depicts object lifelines and chronologically ordered messages exchanged across the UI, service, and database layers during vehicle check-in and check-out.

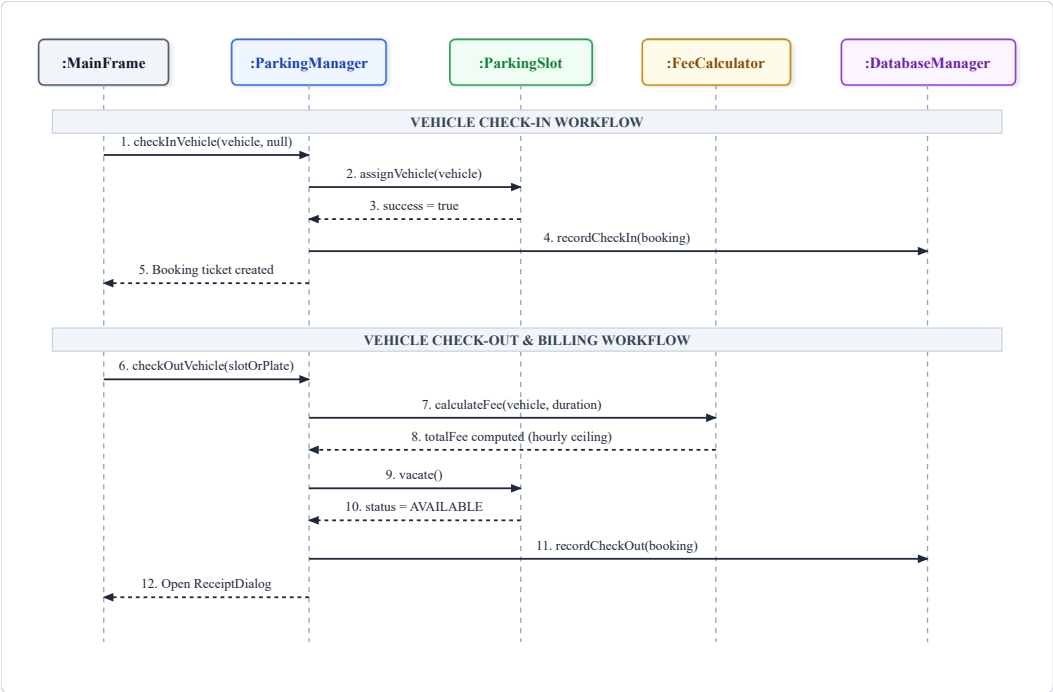


Fig. 4.4: Sequence Diagram — Vehicle Check-In and Check-Out Operations

4.2.3 Activity Diagrams

Activity Diagrams (Fig. 4.5) map the step-by-step procedural control flow, decision branches, and terminal states for check-in and checkout workflows.

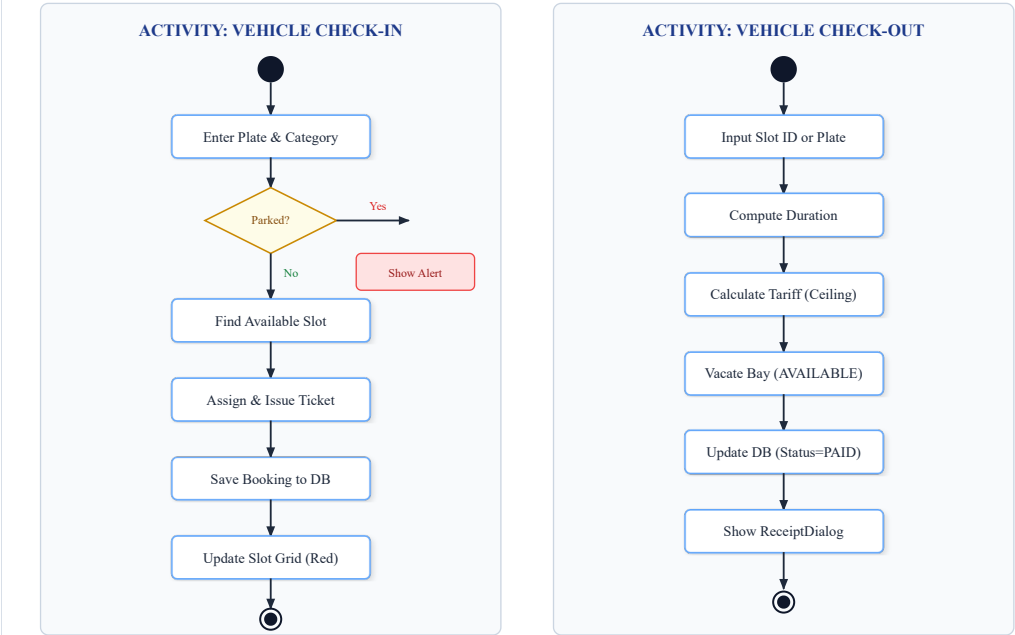


Fig. 4.5: Activity Diagrams — Check-In and Check-Out Process Workflows

4.2.4 Class Diagram

The UML Class Diagram (Fig. 4.6) models the static structural design of the software, showing generalization hierarchies, composition relations, and public contracts.

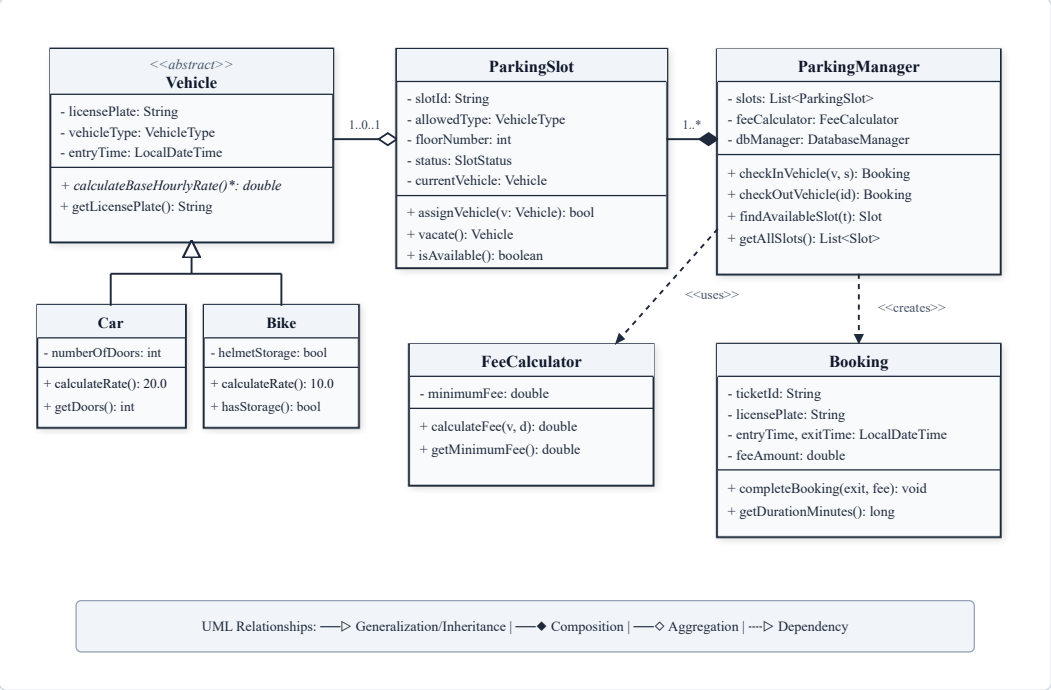


Fig. 4.6: UML Class Diagram — Object-Oriented Architecture of PLMS

4.2.5 Operational Interface Screenshots

The desktop interface provides operators with complete control over bay allocations and transaction histories:

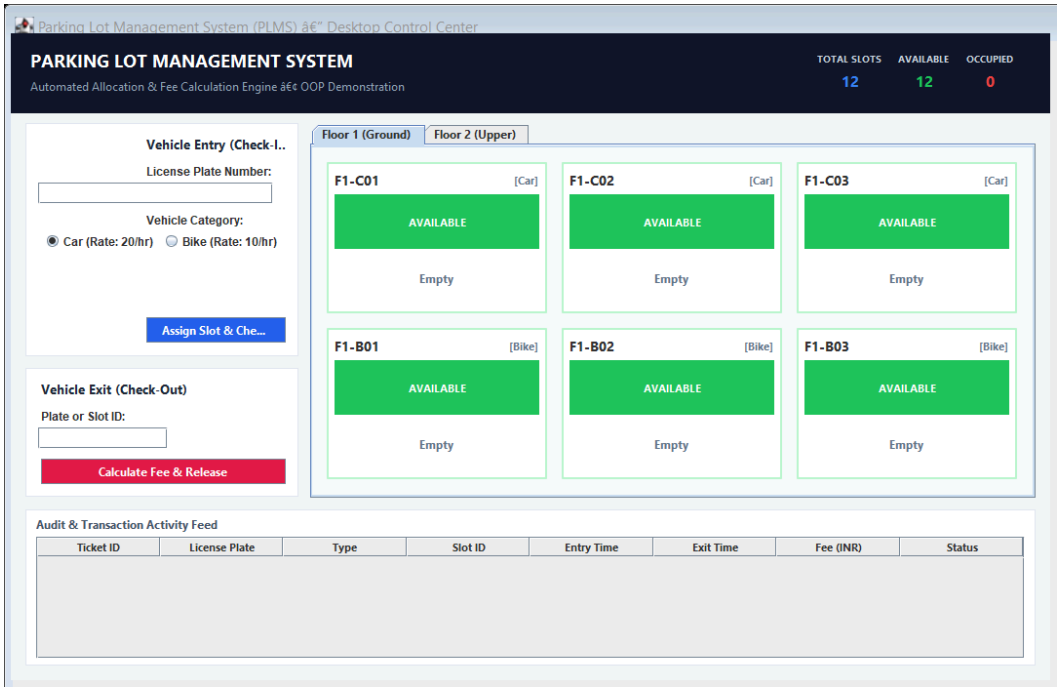


Fig. 4.7: PLMS Desktop Dashboard — Initial Vacant State

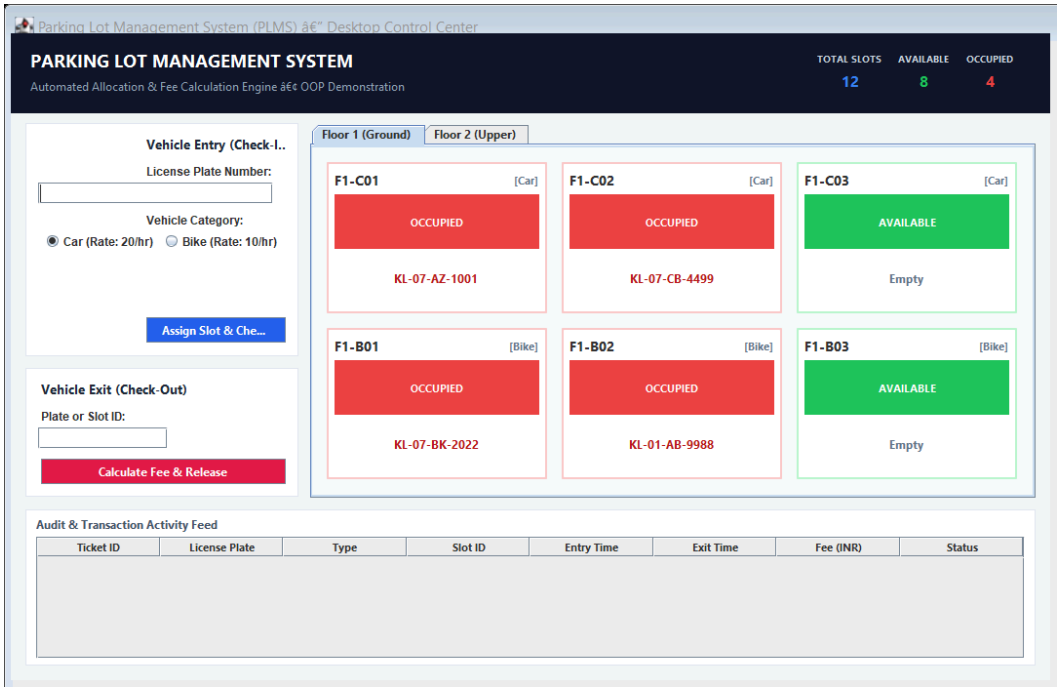


Fig. 4.8: PLMS Desktop Dashboard — Real-Time Multi-Floor Occupancy State

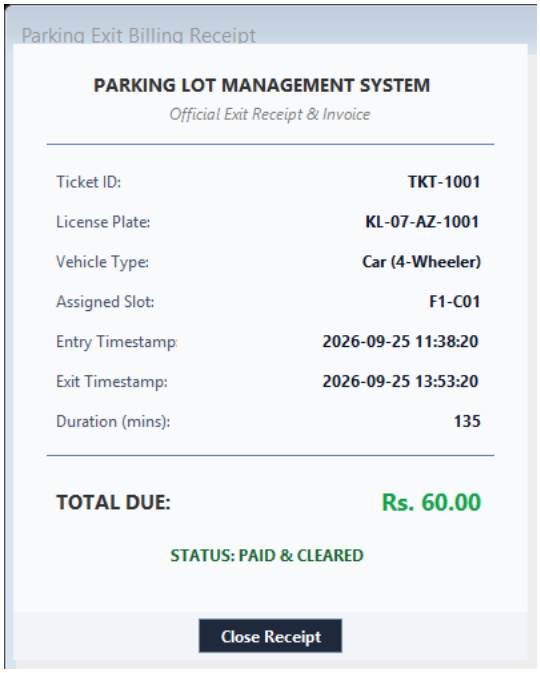


Fig. 4.9: Automated Exit Billing Receipt & Invoice Modal

CHAPTER 5
CONCLUSION

The **Parking Lot Management System (PLMS)** successfully achieves its core objective of delivering a dependable, automated, and object-oriented software solution for vehicular parking management. By leveraging the computational strength of **Java**, the cross-platform GUI capabilities of **Java Swing**, and a robust on-premise relational persistence model, the application guarantees high performance and zero dependency on vulnerable cloud services.

The core strength of the system lies in its deterministic space allocation and automated financial computations. The implementation of dimension-aware slot matching ensures four-wheelers and two-wheelers are routed to appropriately categorized spaces without human confusion. Furthermore, the automated tariff engine, supported by dynamic method dispatch and partial-hour ceiling rules, eliminates manual arithmetic errors and protects revenue integrity.

In addition, the interactive multi-floor dashboard provides attendants with instantaneous situational awareness through high-contrast visual cues, while the synchronized slot mutators prevent race conditions in high-throughput environments. Failsafe CSV audit logging ensures that complete transactional ledgers remain intact even under unexpected hardware restarts or database driver unavailability.

In conclusion, PLMS provides an exemplary demonstration of pure Object-Oriented Programming principles applied to solving practical transportation challenges. The system enhances parking turnaround velocity, eradicates ticket disputes, and establishes a proven foundation for future integrations such as automated camera recognition and microcontroller barrier gates.

CHAPTER 6
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